

10.3

Find $Z_{EQ}(s)$ in Figure P10-3. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros. $R = 4\text{ k}\Omega$, $L = 0.4\text{ H}$, and $C = 100\text{ nF}$.

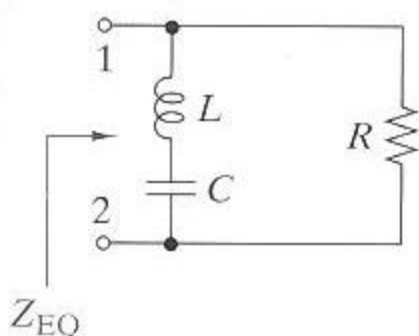


FIGURE P10-3

10-6 Find the poles and zeros of Z_{EQ} in Figure P10-6 for $R = 1\text{ k}\Omega$, $L = 1\text{ H}$, and $C = 500\text{ nF}$.

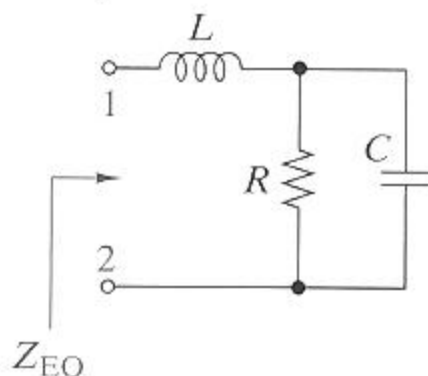


FIGURE P10-6

10-9 Find $Z_{EQ}(s)$ in Figure P10-9. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

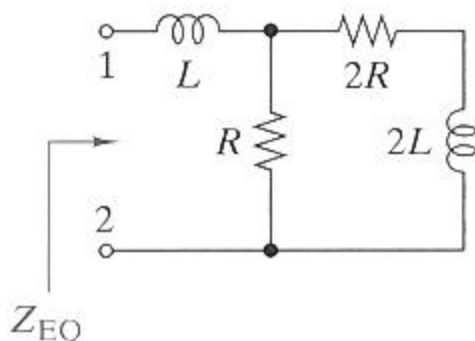


FIGURE P10-9

10-13 The switch in Figure P10-13 has been in position A for a long time and is moved to position B at $t = 0$. Transform the circuit into the s domain and solve for $I_C(s)$ and $i_C(t)$ in symbolic form.

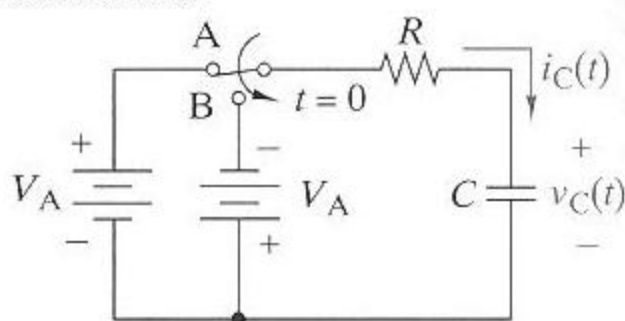


FIGURE P10-13

10-15 There is no initial energy stored in the circuit in Figure P10-15. Transform the circuit into the s domain and find $I_L(s)$ and $i_L(t)$ when $v_1(t) = V_A u(t)$.

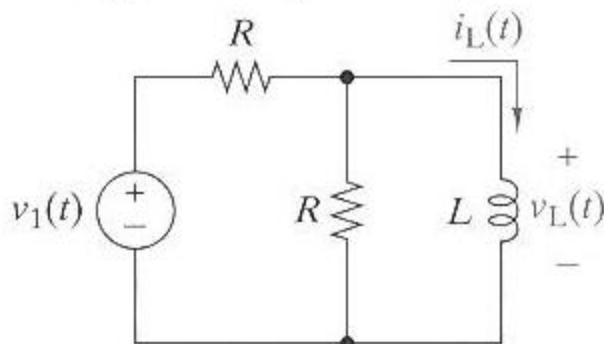


FIGURE P10-15

10-23 There is no energy stored in the capacitor in Figure P10-23 at $t = 0$. Transform the circuit into the s domain and use current division to find $v_O(t)$ when the input is $i_S(t) = 2.5 e^{-100t} u(t)$ mA. Identify the forced and natural poles in $V_O(s)$.

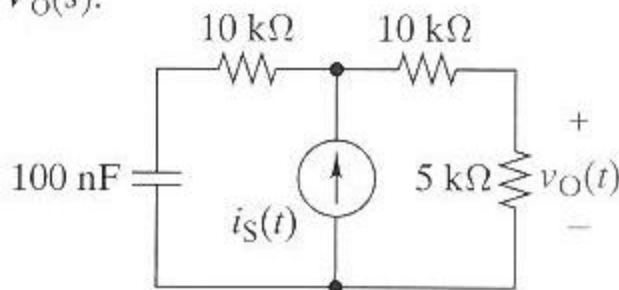


FIGURE P10-23

- 10-25 The circuit in Figure P10-25 is in the zero state. Use a Thévenin equivalent to find the s -domain relationship between the input $I_s(s)$ and the interface current $I(s)$.

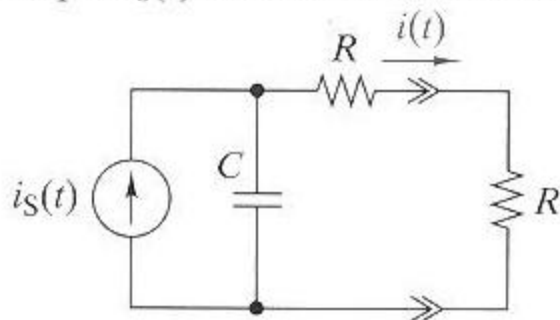
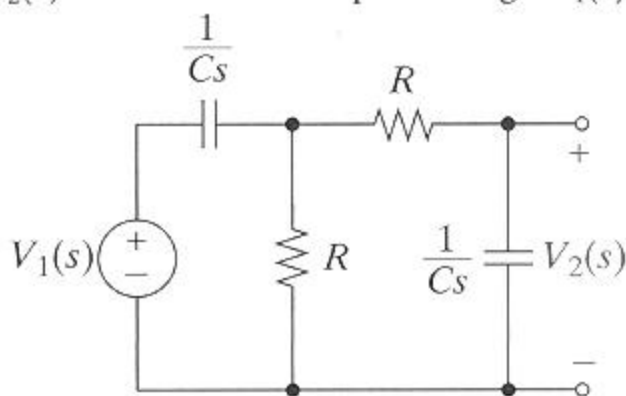


FIGURE P10-25

- 10-26 Repeat Problem 10-25 using a Norton equivalent circuit.



- 10-30 There is no initial energy stored in the circuit in Figure P10-30. Use circuit reduction to find the output voltage $V_2(s)$ in terms of the input voltage $V_1(s)$.



10-33 There is no initial energy stored in the circuit in Figure P10-33.

- Transform the circuit into the s domain and formulate node-voltage equations.
- Solve these equations for $V_2(s)$ in symbolic form.
- Find $v_2(t)$ for $v_1(t) = 30u(t)$ V, $R_1 = 10$ k Ω , $R_2 = 20$ k Ω , $C_1 = 1$ μ F, and $C_2 = 0.5$ μ F.

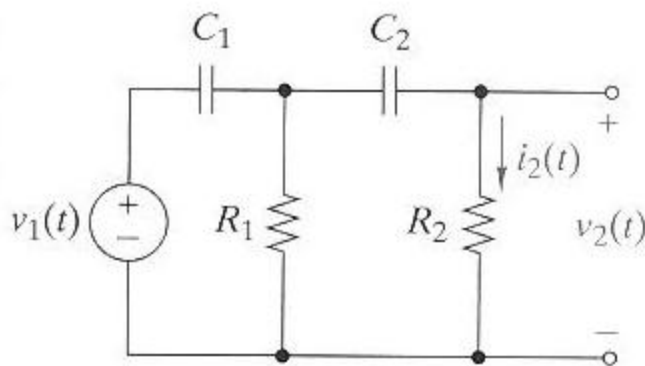


FIGURE P10-33

10-35 There is no initial energy stored in the bridged-T circuit in Figure P10-35.

- Transform the circuit into the s domain and formulate mesh-current equations.
- Use the mesh-current equations to find the s -domain relationship between the input $V_1(s)$ and the output $V_2(s)$.

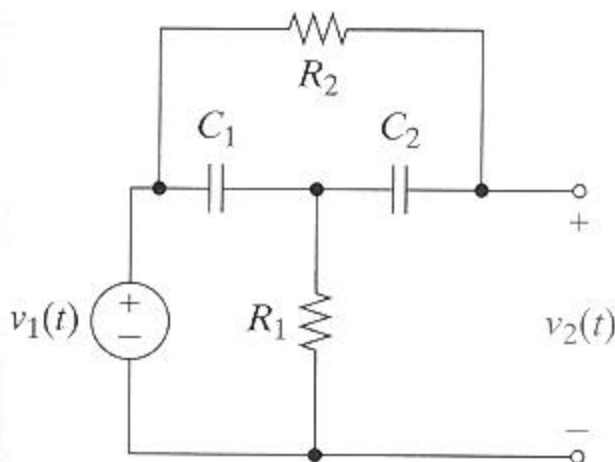


FIGURE P10-35